12. INTRUSION DETECTION

AIM

To understand intrusion detection systems (IDS) and learn basic evasion techniques during cyber attacks.

PROCEDURE

1. Study IDS concepts and types (NIDS and HIDS).
2. Perform reconnaissance using tools like Nmap and observe IDS alerts.
3. Apply evasion techniques (user-agent change, stealth scans).
4. Analyze detection differences and perform post-exploitation activities.

QUESTIONS

TASK 1 - Intrusion Detection Basics

TASK 2 - Network-based IDS (NIDS)

TASK 3 - Reconnaissance and Evasion Basics

Answer the questions below

What scale is used to measure alert severity in Suricata? (*-*)

1-3 ✓ Correct Answer

How many services is nmap able to fully recognise when the service scan (-sV) is performed?

3 ✓ Correct Answer

TASK 4 - Further Reconnaissance Evasion

Answer the questions below

Nikto, should find an interesting path when the first scan is performed, what is it called?

/login ✓ Correct Answer

What value is used to toggle denial of service vectors when using scan tuning (-T) in nikto?

6 ✓ Correct Answer

Which flags are used to modify the request spacing in nikto? Use commas to separate the flags in your answer.

6,A,B ✓ Correct Answer

TASK 5 - Open Source Intelligence

Answer the questions below

What version of Grafana is the server running?

8.2.5 ✓ Correct Answer

What is the ID of the severe CVE that affects this version of Grafana?

CVE-2021-43798 ✓ Correct Answer

If this server was publicly available, What site might have information on its services already?

shodan ✓ Correct Answer

How would we search the site "example.com" for pdf files, using advanced Google search tags?

site:example.com filetype:pdf ✓ Correct Answer

TASK 6 - Rulesets

Answer the questions below

What is the password of the grafana-admin account?

GraphingTheWorld32 ✓ Correct Answer

Is it possible to gain direct access to the server now that the grafana-admin password is known? (yay/nay)

yay ✓ Correct Answer

Are any of the attached IDS able to detect the attack if the file /etc/shadow is requested via the exploit, if so what IDS detected it?

Suricata ✓ Correct Answer

TASK 7 - Host Based IDS (HIDS)

Answer the questions below

What category does Wazuh place HTTP 400 error codes in?

web ✓ Correct Answer

Play around with some post-exploitation tools and commands and make note of what activity is detected by Wazuh; compare it to the activity that's detected by Suricata.

No answer needed ✓ Correct Answer

TASK 8 - Privilege Escalation Recon

Answer the questions below

What tool does linPEAS detect as having a potential escalation vector?

docker ✓ Correct Answer

Is an alert triggered by Wazuh when linPEAS is added to the system, if so what its severity?

5 ✓ Correct Answer

TASK 9 - Performing Privilege Escalation

Answer the questions below

What tool does linPEAS detect as having a potential escalation vector?

docker ✓ Correct Answer

Is an alert triggered by Wazuh when linPEAS is added to the system, if so what its severity?

5 ✓ Correct Answer

RESULT

Understood how IDS detects attacks and how evasion techniques can reduce detection during different attack stages.